Testing for Treatment-Induced Recording Change with Few Clusters

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Abstract

A policy that changes how an outcome is recorded is invisible to pre-treatment diagnostics. This is the position of any difference-in-differences design whose outcome is drawn from an administrative record that the policy, or a reform arriving with it, also rebuilt. The parallel trends deviation reproducing a recording change is exactly zero before treatment, so pre-trend tests reject no more often than their nominal level and sensitivity analysis calibrated from the pre-period loses coverage. Falsification tests across outcome categories graded by recording discretion supply the missing information. Run as one joint test with cluster-robust standard errors, the chi-square calibration is oversized: with 51 clusters, 13 restrictions and 5 categories its null rejection rate is 0.389 against a nominal 0.05. We give three randomization calibrations and characterize the size of each: label permutation, exact in finite samples under category exchangeability and valid for any number of clusters; cluster sign changes, exact under sign symmetry of the cluster contributions; and the restricted wild cluster bootstrap. Label randomization carries a resolution floor that requires at least five categories at the five percent level. Whether a design can detect the recording change its institution implies is computable before the outcome is examined, and it turns on alignment as well as power: the magnitude that matters is the projection of the institutional response on the exposure index, and in all three panels we screen it falls an order of magnitude short of what the design detects.

Keywords: difference-in-differences; measurement error; randomization inference; sensitivity analysis; wild bootstrap.

JEL codes: C12, C15, C18, C23, C81.

1. INTRODUCTION

Applied researchers routinely assess an identifying assumption by asking whether an estimated treatment effect behaves across outcomes the way the assumption predicts, choosing outcomes that differ in their exposure to a suspected violation. For example, Iyer et al. (2012) separate reporting from incidence by contrasting documented crimes against women with crimes against men and gender neutral crimes, and Miller and Segal (2019) run the same contrast for violent victimization by sex of the victim. Weisburd et al. (2024) grade crime categories by their reporting propensity and compare estimated effects across the grades. Baicker et al. (2013) report effects of Medicaid coverage on measured blood pressure and cholesterol alongside self reported diagnoses of the corresponding conditions. In epidemiology the same device is formalized as the negative control outcome (Lipsitch et al., 2010), and Núñez and Matthews (2025) grade candidate controls by how much of the ascertainment process they share with the outcome of interest, which is the cross section this paper uses; Zhang et al. (2025) calibrate a difference-in-differences estimate with negative controls, though against unmeasured confounding rather than a recording response. Run as a single test, the comparison becomes a joint restriction on interaction coefficients, and its p value is read from a chi-square reference distribution with cluster-robust standard errors.

In this paper we study the case in which the suspected violation is a change in the way the outcome is recorded rather than a change in the outcome itself. Campbell and Stanley (1963) named this the instrumentation threat and advised against shifting the measuring instrument and the policy at the same time, advice that is of little use once the policy is what moved the instrument.

How much a recording layer can move is not a small quantity, and it has been measured where the recording technology was assigned at random. Singh (2024) tests over 2,400 schools in Andhra Pradesh and finds that paper-based assessments return scores 16 to 20 percentage points above an independently proctored retest of the same students, while tablet-

based assessments match the retest. The skill being measured is the same in both arms and the instrument is what differs, which is a recording response in the sense of Assumption 1 with the wedge identified by design rather than assumed. That such a change should coincide with a policy is common rather than exotic wherever administrative recording is being rebuilt, which is much of the setting in which quantitative work on developing and emerging economies is done (Singh, 2020). Where the reform to the record arrives with the policy, Assumption 1 is the data generating process and not a hypothetical, and the analyst who reaches for an event-study plot is reaching for the one diagnostic that Corollary 1 says cannot help. We show that for any recording response there exists a structure with no recording response whatever, and with a violation of parallel trends that is exactly zero throughout the pre-treatment period, which generates the same distribution of the observed panel cell by cell. Because the two structures are indistinguishable in the data, no diagnostic computed from the observed panel and the cohort labels rejects more often than its nominal level, whatever the magnitude of the recording response, and pre-trend tests, placebo in time tests and leads and lags plots are all functions of exactly that information.

The same construction tells us what a sensitivity analysis can deliver. The standard response to an untestable assumption is not to test it but to report a confidence set that is honest over a class of deviations (Rambachan and Roth, 2023). In the relative magnitudes class the bound on post-treatment deviations is a fixed multiple of the largest pre-treatment deviation. Under a recording response the quantity being scaled is identically zero, so the class collapses to a single point for every value of its sensitivity parameter and the procedure reports no breakdown value at all. In simulations calibrated to a latent effect of one and a recording wedge of 0.6, and computed with a reference implementation rather than by hand, coverage of the latent effect at a sensitivity parameter of one falls from 0.955 at $N = 400$ to 0.008 at $N = 6{,}400$ and to zero thereafter. An interval whose bound is taken from the recording institution rather than from the pre-period covers at one throughout. More data does not repair the pattern but is instead what produces it, because the calibration input

falls at the sampling rate while the wedge it is meant to bound does not move at all.

Information about a recording response has to come from somewhere other than the pre period. We take it from the cross section of outcome categories, which differ in how much discretion the recording system has over them. That cross sectional comparison is the falsification test with which we began, and the remainder of the paper is concerned with how to calibrate it when the number of clusters is small, which is the situation in which these tests are almost always run. The calibration in current use fails badly at exactly those sample sizes. At 51 clusters, 13 restrictions and 5 categories we measure a null rejection rate of 0.389 against a nominal 0.05, and at 20 clusters the rate is 0.905.

Why it fails has a closed form, and the closed form is what turns the failure into a design rule. When every cluster carries the same design weight, the cluster robust Wald statistic is exactly ω_2 times a Hotelling T^2 with dimension equal to the number of restrictions and degrees of freedom equal to one less than the number of clusters (Hotelling, 1931). The chi-square reference ignores the estimation of the meat matrix from the cluster scores, and the identity prices that omission. Equal weights are what make the statement exact and they hold in no design we run, including both applications, so Proposition 2 earns its place as a diagnostic and not as a calibration we recommend; its own reference distribution rejects between 0.082 and 0.175 across the twelve cells of Table 1. What survives the concession is the rule. The distortion is governed jointly by the number of restrictions and the number of clusters, so the chi-square calibration is valid only when the first grows more slowly than the two thirds power of the second, and a researcher can check that before collecting an outcome. At 36.4 effective clusters size stays below 0.075 only when two or fewer restrictions are imposed.

We then give three calibrations that do not require equal design weights, and we characterize the size of each. The first permutes the exposure labels across outcome categories, and we show that under an exchangeability condition on the joint law of the category data arrays this test has size at most α in finite samples for every number of units, periods,

clusters and categories, with no distributional assumption and no condition whatever on the number of clusters. Its randomization group is the set of relabelings, whose order bounds the resolution of the test, and the smallest attainable p value is $2/C!$ when the statistic is even in the labels and the exposure indices are symmetric about their mean, and $1/C!$ otherwise, so a two-sided test at the five percent level with equally spaced indices requires at least five categories. The second procedure randomizes sign changes over clusters, and it is exactly level α when the cluster contributions are sign symmetric under the null and asymptotically level α when they are not. The third is the restricted wild cluster bootstrap of Cameron et al. (2008), for which we report simulated size rather than a theorem. Both cluster level procedures remain usable when there are too few categories for permutation.

Two further results concern power rather than size. We show that the Wald statistic is not optimal against the alternative these tests are designed to detect, and that when the shape of the response across horizons is known up to scale a directed statistic is uniformly most powerful invariant, which puts the ratio of minimum detectable responses at power 0.80 at 0.393 at 36.4 effective clusters. Label randomization carries a price of its own, requiring 1.35 times the signal at five categories and 1.15 at eight relative to a normal benchmark. We use the label permutation test as our finite-sample benchmark, because its reference distribution is valid for every number of clusters and because it is the only one of the three that holds its size across every design we try. The directed cluster sign-change test is the more powerful procedure and is the one to use when each cluster is a single unit, which is the case in both applications here, but its null rejection rate rises to 0.094 when clusters hold five correlated units and to 0.086 when their sizes are unequal. Its two-sided counterpart is far less affected, at 0.057 and 0.063, so a cluster holding several units is a reason to give up the power of the directed statistic rather than the cluster level calibration.

A falsification test is worth running only against responses it could detect, and whether it can is computable before any outcome is examined. We give that calculation and run it over forty eight specifications of each panel. No specification of either panel is adequate, and

the reason in the crime panel is not a shortage of data. The response the move to incident-based reporting implies is real, averaging 0.011 log points across index categories, but it acts through police counting rules while the exposure index grades victim reporting propensity, so the two are nearly orthogonal and the projection of the one on the other is 0.0044 against a minimum detectable response of 0.052. A design can therefore fail this screen through misalignment rather than through power, and which of the two it is can be read off before the outcome is examined.

The rest of the paper is organized as follows. Section 3 studies what a pre-trend test and a pre-trend calibrated sensitivity analysis can reveal about a recording response. Section 4 states the design and the null. Sections 5 to 8 give the exact null law, the three calibrations and the power results. Section 9 reports size and power, Section 10 screens a design before the outcome is examined, and Section 11 applies the procedures to two panels. Proofs and further evidence are in the online Appendix.

2. RELATED WORK

Our exact null law concerns the cluster robust Wald statistic, which places it in the few cluster literature of Cameron et al. (2008), MacKinnon and Webb (2017) and Carter et al. (2017), and the finite-sample argument is the classical group invariance construction of Lehmann and Romano (2005). Proposition 2 does not introduce the Hotelling family that Bester et al. (2011), Bell and McCaffrey (2002), Tipton and Pustejovsky (2015) and Pustejovsky and Tipton (2018) develop. It states an algebraic identity and, under Assumption 3, an exact law holding at every cluster count and every number of restrictions with $G - m \geq 1$, rather than at matched degrees of freedom or in a large cluster limit. The design that motivates the problem, grading outcome categories by recording discretion, is run by Weisburd et al. (2024); Roth (2022) and Rambachan and Roth (2023) treat falsification tests run as pre-trend tests. The epidemiological statement of that design is Núñez and Matthews (2025), who reach the same grading informally and record what is missing from it: that low precision

complicates reading a negative control, and that no formal guidance exists on how large a response one has to be able to detect. Sections 6 to 10 are an answer to those two sentences. The online Appendix gives the extended discussion.

3. WHAT THE FALSIFICATION TEST IS FOR

That a policy can change how an outcome is recorded rather than the outcome itself is not a new observation. It is Campbell and Stanley (1963)'s *instrumentation* threat to internal validity, and their illustration is almost exactly the design studied below: "the experimental change of putting into office a new principal may produce a change in the recording of discipline infractions rather than in the infraction rate itself." Campbell (1969) makes the same point for policing, where a change in administration reforms the bookkeeping alongside the policy, and we claim no part of that observation for ourselves.

What we do claim is a correction to what the classical tradition concluded about it. For the single series design, Campbell and Stanley (1963) mark instrumentation as the one threat the pre-period cannot dispel, while maturation, testing and regression are ruled out precisely because a pre series is available. Then, for the multiple time series design, which adds a control series and is the ancestor of the design used here, they write that "the difference in the selection of the groups operating in conjunction with maturation, instrumentation, or regression, can hardly account for an apparent effect." Their reasoning concerns instrument differences that arise from how the groups were selected, and a recording response caused by the treatment is not a difference of that kind. No control series can absorb it, because the control units never receive the treatment that induces it. The distinction between an instrument difference across groups and a recording response induced by the treatment is not drawn there, and the applications we cite below suggest that it is not always drawn now.

The consequence we draw is sharper than the classical warning. When the contamination is a recording response, a pre-trend test and a sensitivity analysis calibrated from pre trends are not merely low powered against it, since neither carries any information about it at all.

We state that claim precisely in what follows, price it, and mark the boundary of what it covers.

The gap is a live one, and the online Appendix documents it: work that warns about differential recording and then offers an event-study as the remedy, work that reports a pre-trend test as evidence of parallel trends while conceding that the outcome may reflect disclosure rather than incidence, and work that attributes an undetected violation to low power. Corollary 1 says that where the contamination is a recording response none of these is a matter of power. Where the threat is taken seriously the response is already the method proposed here: Harasztosi and Lindner (2019) settle a reporting story not from the pre-period but across outcomes, which is a cross outcome falsification test conducted informally, and Geruso and Layton (2020) obtain a placebo with power only because an institutional lag supplies a window in which the coding response cannot yet have operated. The rest of this paper supplies the reference distribution such a test needs when the number of clusters is small.

That observation generalizes into the one constructive diagnostic we can offer. The equivalence in Lemma 1 is exact because Assumption 1 places the recording change at the treatment date. Where the recording system responds to an announcement, a rule change or an administrative deadline that precedes implementation, the wedge is nonzero before treatment and a placebo in time test placed in that window has power against it. The same is true of Geruso and Layton (2020)'s lag. We therefore recommend that a researcher who suspects a recording response look for a period in which the policy was already known but the recording system had not yet moved, and run the placebo there. Where no such period exists, no test computed from the outcome panel can help, and that is the case the rest of this paper addresses.

3.1 A recording response

Index units by i, periods by t, outcome categories by c, and adoption cohorts by g, with $\ell_{it} = t - G_i$ the event time and $D_{it} = \mathbf{1}\{t \geq G_i\}$. Write Y^*_{itc} for the latent outcome and Y_{itc} for what the recording system reports.

Assumption 1 (Separable recording response)**.** There exist μ_{ic}, depending arbitrarily on G_i and on the latent outcomes, deterministic ϕ_{tc}, and deterministic paths $m_c(\ell)$ with $m_c(\ell) = 0$ for $\ell < 0$, such that

$$Y_{itc} = Y^*_{itc} + \mu_{ic} + \phi_{tc} + m_c(\ell_{it})D_{it} + u_{itc}, \qquad E[u_{itc} \mid G_i] = 0. \tag{3.1}$$

Two features of Assumption 1 carry the results below and both are restrictions. The distortion enters additively and separably, so a recording map whose response depends on the level of the latent outcome is outside it. And $m_c(\ell) = 0$ for $\ell < 0$ is no anticipation in the recording layer: the recording system changes when treatment arrives, not before. Unit by category terms μ_{ic} are unrestricted, so any time invariant difference in recording practice across units is absorbed and only the treatment-induced change matters.

3.2 Pre trend tests have power equal to size

Let $\Delta_c(g, t)$ denote the deviation from parallel trends in the latent layer. The following is a coupling argument: it exhibits, for any recording response, a structure with no recording response whatever that generates the same observed data.

Lemma 1 (Observational equivalence)**.** *Fix a structure satisfying Assumption 1 with latent parallel trends and recording paths m_c. The structure with no recording response and latent deviation*

$$\Delta_c(g, t) = m_c(t - g)\,\mathbf{1}\{t \geq g\} \tag{3.2}$$

generates the same law of the observed panel, cell by cell. Its pre-treatment deviations are

exactly zero.

Corollary 1 (No pre-treatment diagnostic separates the two)**.** *Let φ be any test that is a measurable function of the observed panel and the cohort labels. Its rejection probability is the same under both structures. In particular a pre-trend test, a placebo in time test and a leads and lags plot reject a recording response with probability no greater than their nominal level, and exactly equal to it when the test is similar on the pre-treatment null.*

The online Appendix answers four objections to Corollary 1: that it restates a familiar fact, that it applies to the test this paper goes on to propose, that pre-treatment parallelism has long been known not to guarantee its post-treatment counterpart, and how the construction differs from the one it resembles.

3.3 Sensitivity analysis calibrated from pre trends

Proposition 1 (The relative magnitudes family is empty against a recording response)**.** *Under Assumption 1 with $m_c \not\equiv 0$, the population identified set for the latent effect under $\Delta^{\mathrm{RM}}(\bar{M})$ is the singleton $\{\vartheta_c(\ell) + m_c(\ell)\}$ for every $\bar{M} \geq 0$, and it excludes the latent effect by exactly the recording wedge.*

Proposition 1 says the relative magnitudes class is empty against a recording response: the class scales a pre-period quantity that Corollary 1 makes identically zero, so it admits no deviation at any value of $\bar{M}$, and the smoothness class calibrated from the pre-period inherits the same failure through its estimated bound. The online Appendix develops both classes and the calibration the reference software uses.

What is prior here should be marked. That a smoothness constant cannot be read off the data while honest coverage is retained is the nonadaptivity result of Low (1997) and Cai and Low (2004), stated for econometric practice by Armstrong and Kolesár (2020) and restated in a difference-in-differences framing by Freeman (2026), and Arora and Chand (2026) note in prose that uninformative pre trends leave the relative magnitudes class with intervals that

fail to widen. Proposition 1 improves on none of that. What it adds is the mechanism: under a recording response the quantity being scaled is zero by construction rather than small by chance, so the collapse is exact, holds at every $\bar{M}$, and is a property of the structure rather than of the realized sample.

3.4 What coverage does

Coverage of the latent effect by a pre-period calibrated honest confidence set falls from 0.955 at $N = 400$ to 0.000 from $N = 25{,}600$ on, while a bound set from the recording institution holds at 1.000 at every sample size and the recorded effect is covered throughout. The calibration input falls at the sampling rate while the wedge does not move, so every relative magnitudes interval eventually excludes the latent effect and larger $\bar{M}$ only delays the crossing. More data makes it worse. The online Appendix gives the full table, the three blocks it contains and the reference implementation the sets are computed with.

3.5 Scope

Three limits bound the claims above. Assumption 1 restricts the recording map to enter additively with a path indexed by event time; a map that responds to the level of the latent outcome is not covered. Lemma 1 and Corollary 1 do not require latent parallel trends, but Proposition 1 is stated for a single adoption cohort, and the extension to staggered designs is not proved here, though the online Appendix reports the same collapse in a three cohort design. And the argument is about what pre-treatment data can reveal; it says nothing against sensitivity analysis whose bound comes from outside the event-study coefficients, which is the practice the coverage experiment of the online Appendix supports and which Armstrong and Kolesár (2020) already recommend.

The last limit is where the result stops rather than where it fails, and it is the one most likely to be mistaken for a refutation. Corollary 1 concerns diagnostics computed from the outcome panel and the cohort labels. It says nothing about identifying a recording response

from data on the recording process itself, and applied work already does this successfully: Ody et al. (2019) separate the Hospital Readmissions Reduction Program from a coincident change in electronic transaction standards using the distribution of diagnosis counts per claim rather than the readmission series. Auxiliary information about the record is not a counterexample to the corollary. It is what the corollary says is required.

What survives is a design problem. If the contamination is a recording response, the pre-period is silent, and information has to come from somewhere else. The remainder of this paper takes it from the cross section of outcome categories, which differ in how much discretion the recording system has over them, and addresses the inference problem that arises when the number of clusters is small.

4. SETUP AND THE NULL

Units $i = 1, \dots, N$ are grouped into clusters $g = 1, \dots, G$; periods are $t = 1, \dots, T$; outcome categories are $c = 1, \dots, C$ with known exposure indices $d_1, \dots, d_C$. Treatment is absorbing with cohort $G_i \in \{1, \dots, T\} \cup \{\infty\}$ and $D_{it} = \mathbf{1}\{t \geq G_i\}$.

Assumption 2 (category balanced design)**.** There is a common cell set $\mathcal{C} \subseteq \{1, \dots, N\} \times \{1, \dots, T\}$ with $|\mathcal{C}| = n_0$ on which every category is observed, and the treatment path D_{it}, the cohort G_i and the cluster map $i \mapsto g(i)$ do not depend on c.

Assumption 2 rules out designs in which categories are measured on different subsamples or at different frequencies. It is verifiable from the data layout. It is a hypothesis of Lemma 2 rather than of Theorem 1: when it fails, $\hat{\gamma}$ in (4.1) is no longer the fixed linear combination of per category event studies that Lemma 2 gives and each category has to be fitted on its own cells, while the randomization of Section 6 acts in the category direction and needs only Assumption 4. Section 11 reports one application in which Assumption 2 fails through a cell pattern that depends on the category label, which breaks Assumption 4 as well.[1]

[1]Fitting each category on its own cells, the label permutation test of (4.2) rejects at 0.047 when the missing cells are tied to the category label and at 0.049 when they are independent of it, against a nomi-

Let L collect the included event time horizons with $|L| = q$, omitting $\ell = -1$ and binning endpoints, let $X \in \mathbb{R}^{n_0 \times q}$ hold the corresponding indicators, and let $P \subseteq \{1, \ldots, q\}$ with $|P| = m$ be the post-treatment block. With y_{itc} the outcome and $\delta_c = d_c - \bar{d}$ the centered exposure index, the pooled gradient regression is

$$y_{itc} = a_{ic} + p_{tc} + \sum_{\ell \in L} \beta_\ell X_{(it),\ell} + \sum_{\ell \in L} \gamma_\ell \delta_c X_{(it),\ell} + u_{itc}, \tag{4.1}$$

estimated by OLS with the CR1 cluster-robust covariance clustered on g. The hypothesis under test throughout is the joint null

$$H_0 : \gamma_P = 0, \tag{4.2}$$

that the post-treatment coefficients do not vary with exposure. This is a weak null on a parameter vector, not a sharp null of no effect for every unit, and the distinction matters for the randomization arguments in Sections 6 and 7: we state separately for each procedure which invariance delivers validity and under what condition.

The cross-category structure enters (4.1) as a single interaction contrast. The procedure tests one joint hypothesis, not C separate hypotheses, so no multiplicity adjustment arises and the error rate controlled is the per comparison rate for (4.2).

Write Π for the two-way within projection on $\mathcal{C}$, $\tilde{X} = \Pi X$, $S = \tilde{X}'\tilde{X}$, and $b_c = S^{-1}\tilde{X}'\Pi y_c$ for the within category event-study coefficient vector.

Lemma 2 (separability)**.** *Under Assumption 2 and* $\operatorname{rank}(\tilde{X}) = q$*, the OLS estimators in* (4.1) *satisfy* $\hat{\beta} = C^{-1} \sum_c b_c$ *and* $\hat{\gamma} = \sum_c \delta_c b_c / \sum_c \delta_c^2$.

The gradient estimator is a fixed linear combination of the per category event studies,

nal 0.05 over 1,200 replications with a Monte Carlo standard error of 0.006 (`verification_extra6_3.json` in `results/inference/`). With the observation rate falling from 1.00 to 0.40 across categories ordered by d_c, the rate is 0.040, against 0.053 for a control that assigns the same five rates to a category drawn at random in each replication, over 1,500 replications with a Monte Carlo standard error of 0.006 (`verification_extra7_8.json`, same directory). The second departure is downward, so the test stays at or below its nominal level and the cost falls on power.

which is what makes the category level randomization of Section 6 available.

The b_c are within category event studies estimated by OLS, which under staggered adoption with heterogeneous effects need not recover a convex combination of cohort specific effects (de Chaisemartin and D'Haultfœuille, 2020; Sun and Abraham, 2021). Replacing each b_c by a heterogeneity robust estimator (Callaway and Sant'Anna, 2021; Borusyak et al., 2024), or by another of the alternatives surveyed in Roth et al. (2023), leaves the randomization arguments of Sections 6 and 7 intact, because those require only that the category level statistic be order invariant in the sense of Theorem 1. The exact null law of Section 5 is specific to the CR1 Wald statistic from (4.1) and would have to be rederived for another estimator.

Failing to reject (4.2) does not establish that the identifying assumption holds. Section 8 gives the minimum detectable response for each design, and the applications in Section 11 report it alongside every nonrejection.

5. THE EXACT NULL LAW OF THE CLUSTER ROBUST WALD STATISTIC

The size distortion has a closed form. We isolate it under an idealization that is used only for this diagnosis and is not maintained elsewhere in the paper.

Let $\tilde{X}_g$ collect the rows of $\tilde{X}$ belonging to cluster g, let u_{gc} be the corresponding errors in (4.1), let $e_{gc} = \tilde{y}_{gc} - \tilde{X}_g b_c$ be the within category residuals, and let $n = Cn_0$ be the number of observations in (4.1). The cluster score of the gradient block and its fitted counterpart are

$$a_g = \frac{\tilde{X}_g' \sum_c \delta_c u_{gc}}{\sum_c \delta_c^2}, \qquad a_g^f = \frac{\tilde{X}_g' \sum_c \delta_c e_{gc}}{\sum_c \delta_c^2}, \qquad \eta_g = S^{-1} a_g, \qquad \eta_g^f = S^{-1} a_g^f, \tag{5.1}$$

all in $\mathbb{R}^q$. Lemma 2 gives $\hat{\gamma} - \gamma = \sum_g \eta_g$, and the CR1 covariance of $\hat{\gamma}$ is $\omega \sum_g \eta_g^f \eta_g^{f\prime}$ with $\omega = \{G/(G-1)\}\{(n-1)/(n-2q)\}$. A subscript P denotes the subvector or submatrix

indexed by the post block, and $\bar{\eta}_P = G^{-1}\sum_g \eta_{gP}$.

Assumption 3 (ideal cluster experiment)**.** (i) Under H_0 in (4.2), the cluster score contributions $\eta_1, \ldots, \eta_G$ are i.i.d. $N(0, \Sigma)$ with $\Sigma_{PP} \succ 0$. (ii) Every cluster carries the same design weight, $S_g = S/G$ for all g, where $S_g = \tilde{X}_g'\tilde{X}_g$.

Part (i) holds when the errors in (4.1) are Gaussian, independent across clusters, and identically distributed across clusters given the design. Part (ii) fails in a staggered design with a large never-treated share; Definition 1 handles that case.

Proposition 2 (exact null law)**.** *Let Assumptions 2 and 3 hold, let* $\operatorname{rank}(\tilde{X}) = q$*, let* H_0 *in* (4.2) *hold, and let* $G - m \geq 1$*. Put* $\omega_2 = (n-2q)/(n-1) = G/\{(G-1)\omega\}$*, let* $\hat{S}_P$ *be the sample covariance of the* η_{gP} *and* $T^2 = G\bar{\eta}_P'\hat{S}_P^{-1}\bar{\eta}_P$*. Then the CR1 Wald statistic* W *for* (4.2) *satisfies* $W = \omega_2 T^2$ *identically, and*

$$\frac{(G-m)\,W}{\omega_2(G-1)m} \sim F(m, G-m) \tag{5.2}$$

exactly for every G *and* m *with* $G - m \geq 1$*. If instead* $G - 1 < m$*, the CR1 meat* $\sum_g \eta_{gP}^f \eta_{gP}^{f\prime}$ *has rank at most* $G-1$*, the Moore–Penrose version of* W *has no Hotelling representation, and* (5.2) *is void.*

The chi-square calibration treats the estimated meat matrix as known. Under Assumption 3 the cost of that substitution is exactly the F denominator in (5.2), which has $G - m$ degrees of freedom. The statement is a degrees of freedom result: nothing about the errors beyond Assumption 3(i) enters, and ω_2 lies between 0.995 and 0.9997 across the designs we study and equals 0.9989 at the headline design of Section 9 ($G = 51$, $C = 5$). The factor $(n-2q)/(n-1)$ counts the $2q$ interacted event time regressors in (4.1) and not the $C(N+T-1)$ absorbed unit by category and period by category effects, so a user calibrating through (5.2) must supply the ω_2 that matches the degrees of freedom convention of their own CR1 implementation.

Inverting (5.2) for a critical value gives a reference distribution that is exact for the null (4.2) under Assumption 3, whose part (ii) requires every cluster to carry the same design weight. Where that part fails the calibration is neither exact nor conservative. In the staggered design of Section 9 with $C = 5$, referring W to (5.2) at the true G gives null rejection rates of 0.174, 0.160, 0.124 and 0.092 at $G = 20$, 30, 51 and 100 against the nominal 0.05, over 2,000 replications per cell (`anatomy_table.csv`, which is also the source for ω_2 above). The version evaluated at the G_{eff} of Definition 1 is an approximation whose accuracy Section 9 measures. The F family in (5.2) is the one that Tipton and Pustejovsky (2015) and Pustejovsky and Tipton (2018) reach by matching denominator degrees of freedom to the data; under Assumption 3 no matching is needed, because (5.2) holds exactly with $G - m$.

Corollary 2 (when the chi-square calibration is valid)**.** *Let* $\text{size}(G, m)$ *denote the null rejection probability of the nominal level-*α *chi-square test under the conditions of Proposition 2. Assume* $G(1 - \omega_2) = O(1)$ *in part (i) and* $\sqrt{m}\,|\log \omega_2| \to 0$ *in part (ii), where* $1 - \omega_2 = (2q - 1)/(n - 1)$.

(i) For fixed m, $\text{size}(G, m) = \alpha + c_m f_m(c_m)\big[(m + c_m)/\{2(G-1)\} - (1 - \omega_2)\big] + O(G^{-2})$ *where* $c_m = \chi^2_{m,1-\alpha}$ *and* f_m *is the* χ^2_m *density, so* $\text{size}(G, m) \to \alpha$ *as* $G \to \infty$. *The correction is positive, and the chi-square test over rejects at all large* G, *when* $2(G-1)(1-\omega_2) < m + c_m$.

(ii) Let $m \to \infty$ *with* $m/G \to 0$ *and* $\chi_{m,G} = \sqrt{m/2}\,\log\{(G-1)/(G-m)\} \to \chi \in [0, \infty]$. *Then* $\text{size}(G, m) \to \Phi(\chi - z_{1-\alpha})$, *and* $\chi_{m,G} = m^{3/2}/(\sqrt{2}\,G) + o(1)$ *whenever* $m^{5/2}/G^2 \to 0$. *If instead* $m/G \to r \in (0, 1)$, *then* $\text{size}(G, m) \to 1$.

Part (ii) is the operative statement. The chi-square calibration is asymptotically valid if and only if $m^{3/2}/G \to 0$, that is $m = o(G^{2/3})$. The online Appendix plots the exact size implied by (5.2) against the number of restrictions at five cluster counts, with the measured rates of Table 1 overlaid.

The limit in Corollary 2(ii) is not accurate at the m this paper works with: along the sequence with $\chi = 0.5$, the exact size from (5.2) exceeds $\Phi(\chi - z_{1-\alpha})$ by 5.95 percentage points at $m = 13$, 4.21 at $m = 25$ and 2.91 at $m = 50$, an error falling like $m^{-1/2}$

(`anatomy_asymptotics.csv`). Every admissible-m figure reported below is computed from (5.2) and not from the limit.

Definition 1 (effective clusters)**.** Let J_P be the $q \times m$ selection matrix whose columns are the standard basis vectors indexed by P, so that $[S^{-1}]_{PP} = J_P' S^{-1} J_P$. For a contrast $w \in \mathbb{R}^m$ put $a = S^{-1} J_P w$ and $\varphi_g = a' S_g a \geq 0$. Then $G_{\text{eff}}(w) = (\sum_g \varphi_g)^2 / \sum_g \varphi_g^2$.

The online Appendix derives G_{eff} under homoskedastic errors independent across clusters and reports the value every design here takes.

6. THE LABEL PERMUTATION TEST

Write $\mathcal{Y}_c \in \mathbb{R}^{n_0}$ for the category-c data array, $\mathcal{D} = (\mathcal{Y}_1, \ldots, \mathcal{Y}_C)$, and $\mathcal{Z}$ for the design, which is common to all categories under Assumption 2. Let $\mathcal{T}(\mathcal{D}, \delta)$ be a statistic with large values evidence against H_0 in (4.2).

The statistic is computed on loading normalized arrays. Let $\hat{\Sigma}_{ck}$ be the sample covariance of the two-way demeaned categories c and k over the untreated cells $\{(i,t) \in \mathcal{C} : D_{it} = 0\}$, and put $L_{ck} = \log|\hat{\Sigma}_{ck}|$. The log triad loading estimate is $\hat{\rho}_c \propto \exp(\hat{a}_c)$, scaled to geometric mean one, with

$$\hat{a}_c = \frac{1}{C-1} \sum_{k \neq c} L_{ck} \; - \; \frac{1}{2(C-1)(C-2)} \sum_{k \neq l,\; k,l \neq c} L_{kl}, \qquad C \geq 3. \tag{6.1}$$

Under a one-factor structure with $\hat{\Sigma}_{ck} \to_p \theta_c \theta_k s^2$ for $c \neq k$, the limit of L_{ck} is $\log|\theta_c| + \log|\theta_k| + \log s^2$, and the two averages in (6.1) isolate $\log|\theta_c|$ up to a constant common to all c, so $\hat{\rho}_c \to_p |\theta_c| / (\prod_j |\theta_j|)^{1/C}$. There is no reference category and no ordering convention in (6.1), which is what Lemma 3 records.

Definition 2 (label permutation test)**.** Normalize each array, $\mathcal{Y}_c^\dagger = \mathcal{Y}_c / \hat{\rho}_c$, and drop every category with $|\hat{\rho}_c| \leq \rho_{\min}$, where $\rho_{\min} = 0.1$ throughout; write C_0 for the count of Section 4, C for the number of categories that survive the screen, and $\mathcal{D}^\dagger$ for the surviving normalized

arrays. For $\pi \in S_C$ let $\delta_\pi = (\delta_{\pi(1)}, \ldots, \delta_{\pi(C)})$ and $T(\pi) = \mathcal{T}(\mathcal{D}^\dagger, \delta_\pi)$. With reference set $\mathcal{R} = S_C$ and $M = C!$, the test rejects at level α when $\hat{p} = M^{-1}\#\{\pi \in \mathcal{R} : T(\pi) \geq T(\mathrm{id})\} \leq \alpha$. When $C! > B$ the reference set is instead $\mathcal{R} = \{\mathrm{id}, \pi_2, \ldots, \pi_B\}$, with $\pi_2, \ldots, \pi_B$ drawn independently and uniformly from S_C, and $M = B$.

The randomization is over the assignments of the exposure labels to the normalized category data arrays, holding the arrays and the design fixed. Nothing is permuted at the unit, cluster or period level. Every run reported below sets $B = 999$, so the reference set is all of S_C at $C \leq 6$ and a Monte Carlo sample at $C = 8$. Enumeration costs $C!$ evaluations of $\mathcal{T}$ and is the binding constraint well before the statistic is: 720 at $C = 6$, 5,040 at $C = 7$ and 40,320 at $C = 8$, against the 999 a Monte Carlo reference set uses at any C. Sampling the group costs resolution rather than validity. Theorem 1 holds on either reference set, the attainable floor becomes $1/B$ by Proposition 3(i), and the Monte Carlo p value has standard error at most $1/(2\sqrt{B})$, which is 0.016 at $B = 999$. The batched implementation evaluates all B label vectors in one pass over the cluster scores, so the cost is linear in B and does not grow with $C!$.

Lemma 3 (equivariance of the normalization)**.** *Under Assumption 2, relabelling the arrays relabels the loading estimates identically: writing $\mathcal{D}_\sigma = (\mathcal{Y}_{\sigma(1)}, \ldots, \mathcal{Y}_{\sigma(C)})$, one has $\hat{\rho}_c(\mathcal{D}_\sigma) = \hat{\rho}_{\sigma(c)}(\mathcal{D})$ for every $\sigma \in S_C$. The retained set $\{c : |\hat{\rho}_c| > \rho_{\min}\}$ is relabeled with them, so $\mathcal{D} \mapsto \mathcal{D}^\dagger$ is S_C-equivariant.*

A statistic that is order invariant in the normalized arrays is therefore order invariant in the raw arrays, and Theorem 1 below applies to $\mathcal{D}^\dagger$ unchanged.

The online Appendix scopes (6.1): two conditions restricting the procedure rather than the panel, and the one-factor structure it inverts, which the crime designs support and the health design does not.

Assumption 4 (category exchangeability)**.** Conditional on $\mathcal{Z}$,

$$(\mathcal{Y}_{\pi(1)}, \ldots, \mathcal{Y}_{\pi(C)}) \stackrel{d}{=} (\mathcal{Y}_1, \ldots, \mathcal{Y}_C) \quad \text{for every } \pi \in S_C,$$

and $d_1, \ldots, d_C$ are fixed constants.

Assumption 4 rules out categories with different error variances and any dependence of the labels on the data. It restricts the category direction only: heteroskedasticity across units and periods, serial correlation, and within cluster dependence of any form are unrestricted, and G is unrestricted. Normalizing the outcome does not restore it under fixed heterogeneous loadings. Dividing category c by its true loading θ_c equalizes the factor component and leaves $\mathrm{Var}(\varepsilon_c/\theta_c) = \sigma^2/\theta_c^2$ category specific, and at $C = 5$ with $\theta = (0.4, 0.7, 1.0, 1.3, 1.6)$ the label permutation test of Definition 2 run on the true loading normalization has size 0.030 against the nominal 0.05, with Monte Carlo standard error 0.006 (`verification.json`, entry `V24`). Exchangeability does hold when the loadings are exchangeable draws independent of d, which is what the simulation of Section 9 imposes, and by Lemma 3 estimating them through (6.1) preserves it. Assumption 4 is not implied by H_0 in (4.2) and it is not directly testable, though its leading failure mode has an observable signature: Section 9 prices the size distortion as a function of the ratio of the largest to smallest category error variance.

Theorem 1 (exact finite-sample validity)**.** *Suppose Assumption 4 holds and $\mathcal{T}$ is order invariant, in the sense that it is a function of the multiset $\{(\delta_c, \mathcal{Y}_c^\dagger)\}_{c=1}^C$ and not of the order in which the pairs are listed. Then for every N, T, G and C, the test of Definition 2 for the null* (4.2) *under the randomization over its reference set $\mathcal{R}$ satisfies* $\Pr(\textit{reject}) \le \alpha$*. If the distinct values among $\{T(\pi) : \pi \in \mathcal{R}\}$ are almost surely attained with a common multiplicity $\mathfrak{m}$, then* $\Pr(\textit{reject}) = \mathfrak{m}\lfloor \alpha M/\mathfrak{m} \rfloor / M$*. Here $\mathfrak{m} = 2$ for the two sided Wald statistic with $\{\delta_c\}$ symmetric about zero, by Proposition 3(ii), and $\mathfrak{m} = 1$ for the one-sided directed statistic. At $C = 5$ and $\alpha = 0.05$ the two give* 0.05 *alike.*

Exactness here is finite-sample exactness for the null (4.2) under the randomization distribution of Definition 2, and it requires Assumption 4. No condition on G, on the error distribution, or on the within cluster dependence structure is used. Order invariance holds for the OLS estimates and CR1 covariance in (4.1), and by Lemma 3 it survives the normalization of Definition 2. It fails when the normalization is not S_C-equivariant, which a

reference category rule keyed to $\arg\min_c d_c$ is not. A position keyed rule that acts as an order invariant rescaling times a scalar common to all categories remains valid, because $\hat{\gamma}$ and its CR1 covariance in (4.1) are homogeneous of degree one and two in the outcome and the Wald statistic is therefore unchanged by a common rescaling: a normalization whose reference column is chosen by the position of the largest permuted label has size 0.052 against the nominal 0.05, with Monte Carlo standard error 0.005 (`verification.json`).

A randomized version has size exactly α. It rejects with probability one when $\hat{p} \leq \alpha_-$ and with probability $(\alpha - \alpha_-)/(\alpha_+ - \alpha_-)$ when $\hat{p} = \alpha_+$, where α_- is the largest attainable value of $\hat{p}$ not exceeding α and α_+ the smallest value exceeding it. No procedure reported below uses it.

Proposition 3 (attainable p values)**.** *(i) On the exhaustive reference set, $\hat{p} \geq |\mathrm{Stab}(\delta)|/C!$, where $\mathrm{Stab}(\delta) = \{\pi \in S_C : \delta_\pi = \delta\}$ has order the product of the factorials of the sizes of the tied groups of $d_1, \ldots, d_C$, so the bound is $1/C!$ when d has no ties. On the Monte Carlo reference set of Definition 2, $\hat{p} \geq 1/B$. (ii) If $\mathcal{T}$ is even in the labels, so that $\mathcal{T}(\mathcal{D}^\dagger, -\delta) = \mathcal{T}(\mathcal{D}^\dagger, \delta)$, and the multiset $\{\delta_c\}$ is symmetric about zero and not identically zero, then $\hat{p} \geq 2|\mathrm{Stab}(\delta)|/C!$ on the exhaustive reference set. Equally spaced d is the leading sufficient case for the symmetry. (iii) On the exhaustive reference set with d untied, a level-α label permutation test therefore requires $C! \geq 2/\alpha$ for a statistic even in the labels with $\{\delta_c\}$ symmetric about zero, that is $C \geq 5$ at $\alpha = 0.05$, and $C! \geq 1/\alpha$ otherwise, that is $C \geq 4$.*

The Wald statistic is even in the labels because $\hat{\gamma}(-\delta) = -\hat{\gamma}(\delta)$ and its covariance estimator is quadratic in the cluster scores. Proposition 3 is a property of the randomization group, not of any implementation. With $C = 3$ the two sided Wald has three attainable p values, the smallest being $1/3$, and the test can never reject at conventional levels. Section 9 reports the resulting rejection rate of exactly 0.000.

7. CLUSTER LEVEL RANDOMIZATION AND THE BOOTSTRAP

Two procedures remain available when $C \leq 4$. Both randomize over clusters rather than category labels, so Proposition 3 does not bind.

The first randomizes sign changes over $\mathcal{E} = \{-1, +1\}^G$. Let γ^r be the restricted estimate, with $\gamma^r_P = 0$ and $\gamma^r_{\text{pre}} = (S_{\text{pre,pre}})^{-1}(S\hat{\gamma})_{\text{pre}}$, let $z^r_g = \sum_c \delta_c \tilde{X}'_g\{\tilde{y}_{gc} - \tilde{X}_g(\hat{\beta} + \delta_c\gamma^r)\}$ be the cluster-g score of the gradient block at that restricted fit, and put

$$\zeta_g = \frac{(S^{-1}z^r_g)_P}{\sum_c \delta_c^2} \in \mathbb{R}^m, \qquad \sum_g \zeta_g = \hat{\gamma}_P. \tag{7.1}$$

The test randomizes $\mathcal{Q}(\epsilon) = (\sum_g \epsilon_g\zeta_g)'\hat{\Omega}^{-1}(\sum_g \epsilon_g\zeta_g)$ over $\epsilon \in \mathcal{E}$, with $\hat{\Omega}$ proportional to $\sum_g \zeta_g\zeta'_g$, and rejects when $\mathcal{Q}(\mathbf{1}) = \hat{\gamma}'_P\hat{\Omega}^{-1}\hat{\gamma}_P$ falls in the upper α tail of $\{\mathcal{Q}(\epsilon)\}$. The CR1 scale factor in $\hat{\Omega}$ is common to every ϵ and cancels from the comparison, and $\hat{\Omega}$ itself does not depend on ϵ because $\epsilon_g^2 = 1$. Write ζ^0_g for the same functional of the true errors under (4.2).

Theorem 2 (cluster sign-change test)**.** *If, under H_0 in* (4.2)*, the cluster contributions are jointly sign symmetric, meaning $(\epsilon_1\zeta_1, \ldots, \epsilon_G\zeta_G) \overset{d}{=} (\zeta_1, \ldots, \zeta_G)$ for every $\epsilon \in \mathcal{E}$, then the sign-change test is exactly level α in finite samples, and so is its Monte Carlo version using $B-1$ random sign vectors together with the identity. Sign symmetry holds for the infeasible statistic built from the true errors under* (4.2) *whenever the cluster error contributions are independent across g and symmetric about zero. Under Assumption 3(ii) the feasible and infeasible contributions coincide exactly on P, so the feasible test is then exactly level α in finite samples. The feasible statistic differs from the infeasible one by $O_p(G^{-1/2})$ and is asymptotically level α as $G \to \infty$ under conditions (W1) to (W3) stated in the online Appendix.*

Under Assumption 3(ii) the gap between the two statistics is zero, because $\zeta_g - \zeta^0_g = -(S^{-1}S_g\gamma^r)_P = -\gamma^r_P/G = 0$ by construction of γ^r, and the measured gap is exactly zero at

$G = 12$, 24, 48, 96 and 192 (`verification.json`). Outside that condition the finite-sample statement applies to the infeasible statistic alone, and the feasible statistic inherits validity only asymptotically in G. The argument for that limit in the online Appendix is a conditional bootstrap central limit theorem rather than an approximate symmetry argument, because the sign-change group has cardinality 2^G. Section 9 measures the finite-sample gap.

The statistic $\mathcal{Q}$ is even in ϵ, so its 2^G values pair and the smallest attainable p value is 2^{1-G} rather than 2^{-G}. An exhaustive enumeration at $G = 7$ gives 64 distinct values out of 128 and a minimum p value of $2/128$ (`verification.json`, entry `E14`). That floor does not bind at any G reported below, the smallest of which is 20.

The second procedure is the restricted wild cluster bootstrap of Cameron et al. (2008), drawing Rademacher weights at the cluster level with (4.2) imposed on the residuals. The implementation partials the unit by category and period by category effects out once and applies the weights to the cluster residuals of the partialled out regression, rather than reabsorbing those effects on every draw; the two versions of the statistic differ by $O_p(G^{-1})$. No finite-sample result is claimed for this procedure. Djogbenou et al. (2019) give conditions for asymptotic validity of the wild cluster bootstrap in a linear regression with clustered errors: independence across clusters, moment conditions on the cluster level scores, and a number of clusters growing without bound with no cluster dominating. Section 9 reports its size in the designs run there and claims nothing beyond that.

MacKinnon and Webb (2017) report simulations in which the cluster robust t-test over rejects more when the $G = 50$ cluster sizes are proportional to United States state populations than when they are balanced, in which the wild cluster bootstrap corrects most of that over rejection, and in which it fails when only a few clusters are treated. Their designs vary the cluster size distribution and the number of treated clusters, not the ratio m/G that drives Corollary 2. Every cluster in the simulation of Section 9 holds one unit observed for $T = 120$ periods, so that simulation cannot test their failure mode either.

The online Appendix sets out the six step recipe for running the three procedures, and

names the file in the replication archive that implements it.

8. POWER, THE MINIMUM DETECTABLE RESPONSE, AND THE OPTIMAL STATISTIC

Let the alternative be $\gamma_\ell = \kappa\, k(\ell)$ for a response path $k(\cdot) \geq 0$ and a scalar magnitude $\kappa > 0$, so that $H_1 : \kappa > 0$. Under local alternatives the noncentrality of the Wald statistic depends on the design through two scalars: $\Sigma_\delta^2 = \sum_c (d_c - \bar{d})^2$, a property of the exposure index, and the timing quantity

$$I_k = k_P' \big([S^{-1}]_{PP} \big)^{-1} k_P, \tag{8.1}$$

where $k_P = (k(\ell))_{\ell \in P}$ is the response shape on the post-treatment block. In the single cohort design with N units, an adopting share p, T_0 omitted reference periods and the constant path $k \equiv \mathbf{1}$, $[S^{-1}]_{PP} = \{Np(1-p)\}^{-1}(I_m + \mathbf{1}\mathbf{1}'/T_0)$ and $I_k = Np(1-p)\, mT_0/(m+T_0)$, which saturates in T: with a single omitted period the information about a level shift never exceeds $Np(1-p)$, however long the panel. The staggered design of Section 9 gives $I_k = 23.62$.

Corollary 3 (closed-form minimum detectable response)**.** *Let* $\lambda^*(\alpha, \beta, m)$ *solve* $\Pr\{\chi^2_m(\lambda^*) > \chi^2_{m,1-\alpha}\} = 1 - \beta$. *With a linear wedge and homoskedastic categories,*

$$\kappa^{\text{Wald}}_{\text{MDE}} = \sigma \sqrt{\frac{\lambda^*(\alpha, \beta, m)}{\Sigma_\delta^2\, I_k}}, \qquad \kappa^{\text{dir}}_{\text{MDE}} = \sigma\, \frac{z_{1-\alpha} + z_\beta}{\sqrt{\Sigma_\delta^2\, I_k}}. \tag{8.2}$$

Both Σ_δ^2 and I_k are computable before any outcome is observed. The third input, σ^2, is the δ^2-weighted mean $\sum_c \delta_c^2 \sigma_c^2 / \sum_c \delta_c^2$ of the category residual variances σ_c^2 of the loading normalized outcome, and it is the only quantity requiring pilot data; a researcher estimates it by the residual variance of the within category regression in (4.1) run on the normalized outcome over untreated cells. Every figure below sets $\beta = 0.20$, so power is $1 - \beta = 0.80$, and $\lambda^*(0.05, 0.20, 13) = 17.85$. The expression for $\kappa^{\text{Wald}}_{\text{MDE}}$ is stated against the chi-square critical value, which Proposition 2 shows is the wrong one; it therefore understates the

minimum detectable response of any correctly calibrated Wald test. Replacing $\chi^2_{m,1-\alpha}$ by the exact critical value from (5.2) raises the required noncentrality from 17.85 to 41.46 at $G_{\text{eff}} = 36.42$ and $m = 13$, inflating the minimum detectable response by a factor of 1.524 (`results/inference/verification.json`, entry `E11`).

At the design of Section 9 with $C = 5$ and $m = 13$, (8.2) returns $\kappa^{\text{Wald}}_{\text{MDE}} = 0.363$ and $\kappa^{\text{dir}}_{\text{MDE}} = 0.214$ for the linear wedge; solving the same noncentrality equation at the nonlinear wedge that design actually generates raises these to 0.450 and 0.242. The simulated magnitude at which size-adjusted power first crosses 0.80 in that design is 0.254 for the directed cluster sign-change test, 0.477 for the directed label permutation test and 0.566 for the chi square Wald at its size-adjusted critical value. Those crossings are interpolated on a coarse κ grid with 600 replications per cell and are accurate to roughly 0.05. Sources: `mde_table.csv` for the closed forms, including I_k and $\lambda^*(0.05, 0.20, 13)$ above, and `perm_mde_vs_sim.csv` for the crossings.

The Wald statistic is not the right statistic against this alternative.

Theorem 3 (optimal directed test)**.** *Consider the Gaussian limit experiment in which the category event studies satisfy $b_c \sim N(\tau + \kappa d_c k, \Sigma)$ independently across c, with $\tau \in \mathbb{R}^q$ a free nuisance and Σ common across categories, so that $\hat{\gamma}_P \sim N(\kappa k_P, V)$ exactly with $V = \text{Var}(\hat{\gamma}_P)$, and suppose V and the response shape k_P are known. Among tests of $H_0 : \kappa = 0$ against $H_1 : \kappa > 0$ that are invariant under the group of shifts $b_c \mapsto b_c + a$, $a \in \mathbb{R}^q$, common to all categories, the test rejecting when $t = k_P' V^{-1} \hat{\gamma}_P / \sqrt{k_P' V^{-1} k_P} > z_{1-\alpha}$ is uniformly most powerful, with power $\Phi(\kappa \sqrt{k_P' V^{-1} k_P} - z_{1-\alpha})$.*

Theorem 3 is stated for known V. In practice V is unknown, and the feasible directed statistic substitutes the CR1 estimate built from G clusters; that is the statistic reported in Table 1 and in the power table of the online Appendix. The substitution costs the normal calibration, and the feasible statistic is the one that the label permutation of Theorem 1 and the cluster sign-change randomization of Theorem 2 calibrate without a normal approximation.

Misspecifying the direction costs power and not level. Running the test at $\tilde{k}$ when the

truth is $\gamma_P = \kappa k_P$ makes $t = \tilde{k}'V^{-1}\hat{\gamma}_P/\sqrt{\tilde{k}'V^{-1}\tilde{k}}$ normal with unit variance and mean $\kappa\cos\vartheta\sqrt{k_P'V^{-1}k_P}$, where ϑ is the angle between $\tilde{k}$ and k_P in the inner product $\langle a,b\rangle = a'V^{-1}b$. Its power is $\Phi(\kappa\cos\vartheta\sqrt{k_P'V^{-1}k_P} - z_{1-\alpha})$, so the minimum detectable response is the $\kappa^{\text{dir}}_{\text{MDE}}$ of (8.2) divided by $\cos\vartheta$, and it is unbounded as $\tilde{k}$ approaches orthogonality to k_P in that metric. The test is level α for every choice of $\tilde{k}$.

The online Appendix works through which statistic the ratio describes, the simulated consequences over 600 replications per cell, the case of a response restricted only to the nonnegative orthant, and how the ratio depends on the calibration of each test.

9. SIMULATIONS

Design. The simulated panel is a two-layer recording model with count outcomes, described in full in the online Appendix and summarized in the note to Table 1. We vary $G \in \{20, 30, 51, 100\}$ with one unit per cluster, $C \in \{3, 5, 8\}$ categories, $m = 13$ post-treatment restrictions and $T = 120$ periods, over 2,000 replications per cell.

Size. Table 1 reports null rejection rates for (4.2) at nominal level 0.05. The chi-square calibration rejects at 0.920, 0.693, 0.393 and 0.172 at $G = 20$, 30, 51 and 100 with $C = 3$. Across C at $G = 51$ the rates are 0.393, 0.389 and 0.356 at $C = 3$, 5 and 8, against 0.378, 0.378 and 0.378 predicted by Proposition 2 evaluated at G_{eff}, which moves in C only through ω_2. The measured decline of 0.037 from $C = 3$ to $C = 8$ runs the other way. The two cells are independently seeded, so the standard error of the difference is 0.015 and the decline is 2.5 standard errors of it. One candidate is the loading normalization, whose estimate improves with C, and we have not isolated it.

The second block of rows refers W/m to $F(m, G-1)$, the usual small-sample patch. Proposition 2 puts the correct denominator degrees of freedom at $G - m$, which is 38 rather than 50 at $G = 51$ and $m = 13$, and the patch removes about a quarter of the excess over the nominal level: 0.309 against 0.389 at $G = 51$, $C = 5$. The Hotelling calibration at G,

Table 1. Null rejection rates, nominal level 0.05

		Clusters G			
Statistic and calibration	C	20	30	51	100
Wald, χ^2_m	3	0.920	0.693	0.393	0.172
	5	0.905	0.686	0.389	0.187
	8	0.917	0.690	0.356	0.172
W/m, $F(m, G-1)$	3	0.838	0.586	0.311	0.143
	5	0.821	0.558	0.309	0.153
	8	0.834	0.571	0.283	0.138
Hotelling at G	3	0.175	0.164	0.122	0.082
	5	0.174	0.160	0.124	0.092
	8	0.172	0.167	0.106	0.088
Hotelling at G_{eff}	3	0.000	0.022	0.047	0.061
	5	0.000	0.020	0.045	0.065
	8	0.000	0.021	0.045	0.055
Label permutation, Wald	3	0.000	0.000	0.000	0.000
	5	0.053	0.050	0.046	0.059
	8	0.051	0.051	0.043	0.058
Label permutation, directed	3	0.000	0.000	0.000	0.000
	5	0.052	0.053	0.048	0.052
	8	0.046	0.050	0.043	0.051
Wild cluster bootstrap, Wald	3	0.053	0.055	0.053	0.061
	5	0.064	0.062	0.051	0.067
	8	0.061	0.061	0.051	0.056
Cluster sign change, Wald	3	0.027	0.036	0.046	0.058
	5	0.022	0.041	0.050	0.066
	8	0.033	0.041	0.048	0.053
Cluster sign change, directed	3	0.058	0.056	0.044	0.045
	5	0.058	0.061	0.057	0.046
	8	0.055	0.055	0.057	0.065

Notes: rejection rates for the joint null $H_0 : \gamma_P = 0$ of (4.2) in the pooled gradient regression (4.1), estimated by OLS with the CR1 covariance clustered on g, at nominal level 0.05. Design: $m = 13$ post-treatment restrictions, $q = 18$ included horizons, $T = 120$, adoption at $t = 48$, 72 and 96 by 20% of units each with 40% never treated, one unit per cluster, exposure indices equally spaced on $[0.1, 1]$. The Hotelling rows refer W to the F law of (5.2) evaluated at G and at G_{eff}, which is $0.727\,G$ at $G = 20$, 30 and 100 and $0.714\,G$ at $G = 51$. Label permutation randomizes the $C!$ assignments of the exposure index to the category data arrays, enumerating all of them at $C = 3$ and $C = 5$ and using 998 sampled assignments together with the identity at $C = 8$. The wild cluster bootstrap uses 499 draws and the cluster sign-change test 999, each including the identity draw. The directed rows use the contrast $w = \mathbf{1}_m$. 2,000 replications per cell; the Monte Carlo standard error is 0.005 at a rate of 0.05, 0.011 at 0.39 and 0.006 at 0.92. Sources: `results/inference/size_table.csv` and `anatomy_table.csv`.

which is the one Proposition 2 licenses under Assumption 3, rejects between 0.082 and 0.175 over the twelve cells. That calibration is model based rather than randomization based, and Assumption 3(i) is what it rests on: with centered χ^2_1 cluster contributions, whose skewness is 2.83, its size is 0.184 (`verification.json`, entry `E4`). We report it as a diagnostic and recommend the randomization calibrations of Sections 6 and 7 in its place.

The two-sided label permutation test is at nominal size for $C \in \{5, 8\}$, ranging from 0.043 to 0.059 over all eight such cells, and its directed version from 0.043 to 0.053. At $C = 3$ both have rejection rate 0.000, which is Proposition 3(iii) rather than a numerical artifact. Three categories is not a rare design, since many recording systems sort outcomes into a handful of coarse classes, and the resolution floor makes label randomization unusable there at any conventional level however large the panel. A researcher with $C = 3$ has three options: use the cluster level procedures of Section 7, which place no requirement on C and carry the sizes reported in the same table; split the outcome into finer categories if the recording system records them separately, since the floor is driven by $C!$ and not by sample size; or report the realized floor alongside the p value so that a nonrejection is not read as evidence. The third is a statement about what the design can support. The wild cluster bootstrap ranges from 0.051 to 0.067. The two-sided cluster sign-change test ranges from 0.022 to 0.066 and is conservative at $G = 20$, where its three rates are 0.027, 0.022 and 0.033; the directed version rejects at 0.058, 0.058 and 0.055 in those same three cells.

The design rule. Applying Corollary 2 through (5.2), the largest number of restrictions holding chi-square size below 0.075 is $m = 2$ at $G_{\text{eff}} = 36.4$, $m = 4$ at 72.7, $m = 6$ at 100 and $m = 8$ at 150. Testing $m = 13$ restrictions at $G_{\text{eff}} = 36.4$ gives a predicted size of 0.379. These are evaluations of (5.2) at values of m that no simulated cell runs; every cell of every block in this section has $m = 13$, so the rule has no simulated counterpart here.

Power. Against a flat recording response of magnitude $\kappa = 0.45$ at $G = 51$ and $C = 5$, the label permutation test with the two sided Wald statistic attains size-adjusted power of

0.383 while the cluster sign-change test applied to the directed statistic attains 0.983, which is the ordering Corollary 3 predicts. The online Appendix gives the power table over the full grid of magnitudes, the contrast used, and the two cells at $C = 5$ where the size-adjusted power of the two-sided permutation test is nonmonotone in κ.

Further evidence. The online Appendix reports seven further blocks of this design: the exact null law checked in an experiment built to satisfy Assumption 3, the accuracy of the substitution of G_{eff}, the size distortion when category exchangeability fails and what studentizing does to it, the price label randomization charges relative to a normal benchmark, how the exposure spread, the panel length and the pre window move the minimum detectable response, the behavior of the calibrations in four harder designs with clustered, unequal, serially correlated and heavy tailed errors, and a benchmark against the bias-reduced variance of Bell and McCaffrey (2002). None of them changes a conclusion drawn here.

10. WHAT THE DESIGN HAS TO SATISFY

A falsification test is worth running only against responses it could detect, and whether it can is a property of the design that is computable before the outcome is examined. Write κ_{MDE} for the response the design detects with probability 0.80 at level α, which by Corollary 3 depends on the number of restrictions, the path information I_k, the exposure spread Σ_δ^2 and the residual variance.

What κ_{MDE} has to be compared with needs care, because the test does not see the whole of a recording response. Let m_c be the log response the recording change produces in category c. The alternative the gradient test is built against has cross-category shape proportional to δ_c, so the part of m that the test can detect is its projection on the exposure index,

$$\kappa^{\text{inst}} = \frac{\sum_c \delta_c m_c}{\sum_c \delta_c^2}, \tag{10.1}$$

and the design is adequate when $\kappa_{\text{MDE}} \leq \kappa^{\text{inst}}$. Both inputs are external to the outcome

panel, d_c from the recording institution and m_c from whatever is known about the recording change, so (10.1) is computable in advance, which is the only reason the comparison is worth making. A quantity read off the post-treatment estimates, a per category breakdown value for instance, is not a candidate for κ^{inst}: it would make the screen a function of the outcome it screens.

Five features of the specification move the ratio.

The statistic is the first. By Corollary 3 the directed statistic has the smaller minimum detectable response whenever the response path is signed, which the recording mechanism makes it. The pre window is the second, and Corollary 1 is what makes narrowing it free: the pre-period carries no information about a recording response, so horizons spent there buy nothing this test can use while competing with the post block for identification. Dropping the crime pre window from six horizons to one raises I_k from 40.5 to 175.5 and cuts κ_{MDE} by the square root of that. The number of restrictions is the third, through the critical value. The fourth is the exposure spread, and the cheapest way to widen it is a category the recording system has no discretion over, which anchors the low end of d at zero.

The fifth is alignment, and it is the one that can defeat the other four. By (10.1) the numerator is an inner product, so a design can have ample power against a response of the average size the institution implies and almost none against the response it actually implies, if the index orders the categories on a different principle from the one the recording system acts on. The cosine between δ and m is the whole of it, and it costs one inner product to check. Section 11 reports a design that fails here while passing on every other lever, which is the case a researcher most needs to be warned about, because widening the spread and narrowing the pre window both look like progress and neither helps.

11. APPLICATION

We apply the procedures to two state panels assembled for the study of treatment-induced recording change. The identification claim in each case is that under measurement invariance

the post-treatment event-study coefficients do not vary with the exposure index, so (4.2) holds.

The first four levers of Section 10 move the crime panel a long way. Homicide, which the panel already carries and which the specification as originally run holds out as a $d = 0$ series, more than doubles Σ_δ^2 from 0.349 to 0.769 when it enters the gradient; dropping the pre window from six horizons to one raises I_k from 40.5 to 175.5; and the directed statistic takes the rest. Over the forty eight specifications we enumerate the minimum detectable response falls to 0.052 at its best, with $C = 7$, $m = 9$ and $G_{\text{eff}} = 42.9$.

What it has to beat is far smaller. The FBI's comparison of the two reporting systems over 6,299 agencies puts the log change in recorded counts at under 0.001 for rape, 0.006 for robbery and for aggravated assault, 0.010 for burglary, 0.026 for larceny and 0.027 for motor vehicle theft, with murder unchanged, averaging 0.011 across the seven index categories (Congressional Research Service report R46668, which also reports the earlier Bureau of Justice Statistics comparison over 1,131 agencies). Projected on the exposure index through (10.1) the first gives $\kappa^{\text{inst}} = 0.0044$ and the second 0.0017. No specification of the crime panel is adequate, the best ratio being 11.8.

The shortfall is misalignment rather than power, and that distinction is the useful part. The responses themselves are of the order the design detects; it is their projection on δ that is not, the correlation between the two being 0.14 under the first comparison and 0.03 under the second. The exposure index grades categories by the share of victimizations not reported to police, and the move to incident-based reporting does not act on victim reporting at all. It drops the hierarchy rule and makes co-occurring offenses countable, which is a change in police counting, and the two orderings come apart and in places reverse: motor vehicle theft has the second lowest d_c and nearly the largest response, because vehicle theft often accompanies another index offense, while rape has nearly the highest d_c and no response worth measuring, because it rarely accompanies a more serious one. Widening the spread and narrowing the pre window both looked like progress here and neither helped.

The same calculation screens the health panels out, the best ratios being 12.5 in the annual sub design and 14.8 in the odd-year sub design against 33.9 and 53.2 as run. The exposure index there already places three conditions at $d = 0$, so widening the spread has little left to buy, and the shortfall is again an order of magnitude.

We therefore report all three designs as illustrations of the calculus screening a design out, and none of them as evidence about incident-based reporting or about Medicaid expansion. Across every specification of both panels the randomization calibrations reject at about their nominal rate, ten of the two hundred and forty three crime p values and nineteen of the three hundred and seventy eight health ones falling at or below 0.05, which is what the null predicts and what a design this far from adequate can be expected to produce either way. Those nonrejections carry no information about measurement invariance. The point of computing κ_{MDE} and κ^{inst} before the outcome is that all of this was knowable in advance. Sources: `design_power.csv` for the screen and `theory/institutional_wedge.md` for the derivation of κ^{inst}.

The online Appendix describes the two panels, the loading normalization applied to each category array, and what the chi-square calibration returns on them in contrast to the randomization calibrations.

Data availability. The recorded crime series are the FBI's summary based estimated state totals; the NIBRS adoption dates come from the FBI Crime Data Explorer agency API; the reporting rates defining d_c come from Bureau of Justice Statistics, *Criminal Victimization 2019*, NCJ 255113, Table 6. The health series are BRFSS crude prevalence from the CDC Socrata dataset `dttw-5yxu`, and the expansion effective dates come from the Kaiser Family Foundation table of state Medicaid expansion decisions. All are public. The two constructed panels and the code that builds them are in the replication archive, whose location is withheld for anonymous review, under `data/processed/` and `scripts/`. The per category changes in recorded counts underlying κ^{inst} are from Congressional Research Service report R46668,

which summarizes the FBI and Bureau of Justice Statistics comparisons of the two reporting systems; the projection is derived in `theory/institutional_wedge.md`. Every p value, G_{eff} and predicted size is read from `results/empirics/inference_gradient.csv` and every realized resolution floor from `results/empirics/app_floors_injection.csv`. The forty eight screened specifications of each panel, with their minimum detectable responses and adequacy ratios, are in `results/inference/design_power.csv`.

12. CONCLUSION

We have argued that a treatment-induced change in recording cannot be separated from a treatment effect by anything computed from the outcome panel. Under Assumption 1 the deviation that reproduces such a change is exactly zero before treatment, so no diagnostic of that kind rejects more often than its nominal level, and the relative magnitudes class scales a pre-period quantity that is identically zero and therefore admits no deviation at any value of $\bar{M}$. Its coverage of the latent effect falls from 0.955 to 0.000 as the panel grows, while a bound set from the recording institution instead of from the pre-period holds at 1.000 throughout.

Information about such a change must come from the cross section of outcome categories, and the test that uses it needs a reference distribution of its own. We have shown that the chi-square distribution is badly oversized at the cluster counts these tests are run on, since at $G = 51$ and $m = 13$ its size is 0.389. We therefore give three alternatives and characterize each. The label permutation test of Definition 2 is exact in finite samples for (4.2) under Assumption 4 and requires no condition on G, the cluster sign-change test is exact under sign symmetry of the cluster contributions, and the bootstrap carries no exactness claim. Their simulated sizes in Table 1 are 0.043–0.059, 0.022–0.066 and 0.051–0.067 at $C \in \{5, 8\}$.

Three limitations bound what we claim. Proposition 1 is proved only for a single adoption cohort, where the online Appendix shows by simulation that the conclusion survives staggering. Proposition 2 requires equal design weights, which no design we study satisfies,

so we use it to explain the distortion rather than to replace it. Assumption 4 fails when categories have different error variances: in a Gaussian limit experiment that failure is liberal, at size 0.102 for a variance ratio of 25 at $C = 5$, while in the panel, where the CR1 covariance is recomputed for every permuted label vector, it is conservative instead and costs power. The cluster level procedures do not rely on the assumption at all.

Lemma 2 requires a category balanced design, a restriction the health panel shows to be binding, and what blocks Theorem 1 there is Assumption 4, because the pattern of observed cells depends on the category label. Extending the reduction to unbalanced panels would recover the procedure in settings where categories are surveyed on different schedules, and we leave that extension to future work.

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Funding. No funding was received for conducting this study.

Competing interests. The author has no competing interests to declare that are relevant to the content of this article.

Ethics approval. Not applicable. The study uses only aggregated public use statistics and involves no human participants, their data or biological material.

Data, material and code availability. Every series used here is a public use file. The sources, the two constructed panels and the code that builds them are listed under data availability in Section 11, together with the replication archive that holds them and the output file behind every number reported in this paper.

Use of generative AI. The author used a generative AI assistant during preparation of this manuscript, for drafting and editing prose and formatting the reference list, and for some of the simulation code. This goes beyond AI assisted copy editing and is declared here

for that reason. All theoretical results and their proofs, the design of the empirical analysis, and the interpretation of all findings are the author's own. The author has verified every reported number against the output files named in the notes to the tables and takes full responsibility for the content.

Online Resource. The online Appendix referred to throughout contains every proof and the supporting simulation and application evidence. It restates nothing and follows the numbering used here, and is submitted as Online Resource 1.

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